INCH-POUND
MIL-DTL-24576B
4 November 2024
SUPERSEDING
MIL-C-24576A
27 July 1987

DETAIL SPECIFICATION

CLOTH, SILICA GLASS; CLOTH COATED, GLASS, SILICONE RUBBER COATED

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification establishes the requirements silica glass cloth and silicone rubber coated glass cloth.

1.2 Classification. The cloth will be in the following types and classes.

1.2.1 Types. The cloth will be available in the following types.

Type I – Silica glass
Type II – DELETED
Type III – Fibrous glass coated with silicone rubber

1.2.2 Classes. The cloth will be available in the following classes.

Class 1 – Heavyweight
Class 2 – Lightweight

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AMSC N/A FSC 8305

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents be those listed in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-Y-1140	- Yarn, Cord, Sleeving, Cloth, and Tape-Glass
MIL-DTL-32075	- Label: For Clothing, Equipage, and Tentage, (General Use)
ASSIST	- https://assist.dla.mil

2.3 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U.S. COAST GUARD (USCG)

46 CFR Subpart USCG 164.009 - Noncombustible Materials for Merchant Vessels

(Copies of this document are available online at https://www.govinfo.gov).

2.4 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMIST AND COLORISTS (AATCC)

AATCC TM20A - Fiber Analysis: Quantitative

(Copies of this document are available online at https://www.aatcc.org.)

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ASTM INTERNATIONAL

ASTM C871	- Standard Methods for Chemical Analysis of Thermal Insulation Materials for Leachable Chloride, Fluoride Silicate, and Sodium Ions
ASTM D512	- Standard Test Methods for Coated Fabrics
ASTM D751	- Standard Test Method for Warp (End) and Filling (Pick) Count of Woven Fabrics
ASTM D1776/D1776M	- Standard Practice for Conditioning and Testing Textiles
ASTM D1777	- Standard Method for Measuring Thickness of Textile Materials
ASTM D2000	- Standard Classification System for Rubber Products in Automotive Applications
ASTM D3773/D3773M	- Standard Test Methods for Length of Woven Fabric
ASTM D3774	- Standard Test Methods for Width of Woven Fabric
ASTM D3775	- Standard Test Method for Fabric Count of Woven Fabric
ASTM D3776/D3776M	- Standard Teat Methods for Mass per Unit Area (Weight) of Woven Fabric
ASTM D3990	- Standard Terminology Relating to Fabric Defects
ASTM D4029/D4029M	- Standard Specification for Finished Woven Glass Fabrics
ASTM D4157	- Standard Test Method for Abrasion Resistance of Textile Fabric (Oscillatory Cylinder Method)
ASTM D5035	- Standard Test Method for Breaking Force and Elongation Textile Fabrics (Strip Method)
ASTM D6413/D6413M	- Standard Test Methods for Flame Resistance of Textiles (Vertical Test)

(Copies of these documents are available online at https://www.astm.org.)

2.5 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence unless otherwise noted. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2).

3.2 Cloth. The cloth shall be asbestos free. Asbestos and components containing asbestos shall be prohibited.

3.2.1 Type I cloth. The warp and fill yarns of Type I cloth shall be composed of amorphous silica glass. The cloth shall not contain additive amount of cristobalite, quartz, and tridymite. Significant amounts of these minerals shall not be added during processing (see 4.4.4).

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3.2.2 Type III cloth. The warp and fill yarns used in the manufacture of type III base cloth shall be composed of 100 percent continuous filament glass yarns and shall be in accordance with MIL-Y-1140.

3.2.3 Physical requirement, uncoated and untreated cloth. Prior to coating or treatment, the cloth shall be as specified in Table I.

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TABLE I. Requirements for uncoated or untreated cloth

Property	Type I		Type III
	Class 1 Heavyweight	Class 2 Lightweight	
Silica content, percent, minimum	96	96	--
Weave	8 or 12 Harness Satin	8 Harness Satin	8 Harness Satin
Style 1/	--	--	Style no 1584
Yarns per inch			
Warp	42-54	42-54	42-46
Fill	32-44	32-44	33-37
Weight (oz/yd^2)	31-39	17-21	23-29
Color	Tan	Tan	White/Buff

1/ Standard commercial style designation.

3.2.4 Physical requirement coated or treated cloth. The coated or treated cloth shall be as specified in Table II.

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TABLE II. Requirements for treated or coated cloth

Property	Type I		Type III
	Heavyweight	Lightweight	
Weight (oz/yd^2)	31-39	17-21	30-36
Thickness (inch)	0.040-0.062	0.023-0.032	0.035-0.043
Color	Red / Black	Red / Black	Red / Black
Breaking strength			
Warp (min)	110	80	375
Filling (min)	40	55	225
Flame Resistance			
After flame (max, seconds)	1/	1/	5
After glow (max, seconds)			4
Char length (max, inch)			1
Abrasion resistance (cycles)			
Warp (min)	100	20	800
Fill (min)	60	20	800
Blocking rating (max)	None Required	None Required	No. 1
Chloride content ppm (max)	250	250	250

1/ The flame resistance requirements shall meet those specified under 46 CRF Subpart USCG 164.009

3.3 Treatment.

3.3.1 Type I treatment. Type I cloth shall be impregnated from both sides with a treatment to minimize dusting and to increase the abrasion resistance of the cloth. The treatment shall contain no fluorocarbons or any other halogenated hydrocarbons.

3.3.2 Type III treatment. Type III cloth shall be impregnated on both sides with a silicone rubber compound conforming to ASTM D2000, M3FC 507 B37.

3.4 Pigment and color. The color of the Type I or Type III cloths shall be specified in the contract. When color is not specified there shall be no added pigmentation. (see 4.4.7)

3.4.1 Nonpigmented cloth. When no pigment is added, the Type I cloth shall be tan, and the Type III cloth shall be white/buff.

3.4.2 Pigmented cloth. When a red cloth is required, it shall be pigmented with iron oxide to produce the red color. When a black cloth is required, is shall be pigmented with zinc oxide to produce the black color. The coloring agent shall be flame resistant as specified in 3.2.4.

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3.5 Non-fibrous material. The total sizing, finishing and other non-fibrous material content shall not exceed 5 percent when tested as specified in 4.4.1.

3.6 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.7 Width. The width of Type I cloth, including selvage, shall be 36 (+2, -1) inches. The width for Type III cloth shall be 38 or 60 (+2, -1) inches. (see 4.4.8).

3.8 Length. Unless otherwise specified, the cloth shall be provided in rolls not less than 50 yards, nor more than 55 yards, in length. Rolls shall contain no more than two pieces, and no individual piece shall be less than 5 yards. (see 4.4.12)

3.8.1 Cloth put-up. Each length of cloth furnished shall be put upon rolls, full width. (see 4.4.9)

3.9 Fiber identification label. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with MIL-DTL-32075, Type VII. (see 4.4.11). The information below shall be cited on the label or ticket. As an alternate to the term “Non-Asbestos Cloth”, the term “Asbestos Free Cloth” may be used.

Nomenclature: Cloth, Glass (example) NSN:
8305-00-000-0000 (example)
Specification Number, Type, Class: MIL-DTL-24576, Type I, Class 2 (example) Contract Number: SPE1C1-24-D-0000 (example)
Fiber Content: 100% Glass (example) Length: 53
yards (example)
NON-ASBESTOS CLOTH

3.9.1 Cloth markings. Unless otherwise specified, one side of the glass cloth shall be marked or stenciled in accordance with MIL-DTL-32075, Type IV with the information below. The markings shall be not less than 3/8 inch in height. The marking shall be on every linear yard of the cloth.

NON-ABESTOS CLOTH
Specification number and class
Manufacturer’s name Manufacturer’s
product identity.

3.10 Noncombustible. Type I material shall be noncombustible (see 4.4.1).

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. When a first article inspection is required, the inspection shall consist of the examinations and tests specified in 4.4.1 through 4.4.12 (see 3.1).

4.2.1 First article sample. Unless otherwise specified in the contract (see 6.2), the first article sample shall consist of a 3-yard, full-width sample of each type and class of cloth provided at any one time.

4.3 Conformance inspection. Conformance testing shall consist of all examinations and tests specified in 3.2 through 3.11.

4.3.1 Conformance inspection sampling. The sample unit shall be (1) continuous three (3) yard length of full width fabric per roll of finished cloth, except for the shade evaluation cited under 4.4.5 shall not apply. The number of samples per lot size shall be a specified below. Unless otherwise specified in the contract, conformance verification for the coated cloth shall be performed on the sample roll sizes below.

Lot Size (yards)	Sample Size (rolls)
800 or less	2
801 up to including 22,000	3
22,001 and over	5

4.3.1.1 In-process inspection. Visual and dimensional examinations of the cloth shall be made at any point or during any phase of the manufacturing process to determine whether manufacturing details, which cannot be examined in the finished product, are in accordance with requirements specified in Section 3. Materials and components, which can be classified, as a defect in accordance with ASTM D3990 for woven cloth and Section 4 shall be removed from production.

4.4 Inspection methods.

4.4.1 Finished cloth tests. The cloth shall be tested as specified in Table II. The finished cloth shall be conditioned as directed in ASTM D1776/D1776M. All test reports shall contain the individual values utilized in expressing the results. (see 3.2.3, 3,2,4). The verification requirements for the uncoated cloth will be specified in the contract.

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TABLE III. Finished cloth tests

Characteristics	Requirement	Verification
Silica content	3.2.3	4.4.4
Weave	3.2.3	Visual 1/
Style	3.2.3	ASTM D4029
Non-fibrous	3.5	AATCC 20A
Yarns per inch	3.2.3	ASTM D3775
Treatment	3.3	4.4.2
Pigment	3.4	2/
Weight	3.2.3	ASTM D3776/ASTM D3776M
Thickness	3.2.4	ASTM D1777
Width	3.7	ASTM D3774
Length	3.8	ASTM D3773
Color	3.4	Visual
Breaking strength	3.2.4	ASTM D5035
lame resistance, Type III After flame After glow Char length	3.2.4	ASTM D6413/D6413M
Abrasion resistance	3.2.4	4.4.2
Blocking	3.2.4	4.4.3
Chloride content	3.2.4	3/
Noncombustibility	3.10	USCG 164.009 4/

1/One 6 x 6-inch test specimen shall be evaluated for weave. The results shall be returned as pass or fail.
2/ One 6 x 6-inch test specimen shall be evaluated for the presence of zinc oxide for the color black and iron oxide for the color red. Any test methodology may be used. The results shall be recorded as pass or fail.
3/ Preparation of the sample shall be in accordance with ASTM C871. Testing shall be in accordance with ASTM D512, method A. The results shall be recorded as pass or fail.
4/ Subparts of 46 CFR 164.009-1, General; 164.009-7, Contents of Application, and 164.009-9 Procedure for Approval shall apply.

4.4.2 Abrasion resistance. The abrasion resistance test shall be performed as specified in ASTM D4157, except that the test shall be continued until destruction of the cloth occurs. For the testing of Type I cloth, a 2-pound load, 2-pound tension, and a 600-grit paper shall be used; for the testing of Type III cloth, a 2-pound load, a 6-pound tension, and 320-grit paper shall be used. During testing of all types, the yarns facing the drum shall be parallel to the direction of oscillation (see 3.2.4).

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4.4.3 Blocking test. The blocking of the fabric shall be determined by following ASTM D751. The test shall be performed at a temperature of 82 ± 2 °C for 30 minutes. Evaluate the resistance of the specimen to blocking by the scale give below:

1 – No blocking. Cloth surfaces are free and separate without any evidence of cohesion or adhesion.
2– Trace blocking. Cloth surfaces show slight cohesion or adhesion.
3 – Slight blocking. Cloth surfaces show slight cohesion or adhesion.
4 – Slight blocking. Sloth surfaces must be lightly peeled to separate.
5 – Blocking. Cloth surfaces separate with difficult, or coating is removed during separation.

4.4.4 Silica content of cloth. One test specimen shall be evaluated per lot. The silica content of the cloth shall be determined as follows:
(a) Place the specimen in a crucible and ignite with a burner under full flame for 10 minutes.
(b) Cool in a desiccator until ambient temperature equilibrium is established and weigh a 0.5 ± 0.1-gram specimen to 0.001 gram into a pre-weighed platinum crucible with cover.
(c) Add five drops of 1:1 sulfuric acid-water solution.
(d) Carefully add enough 49% reagent grade hydrofluoric acid to fill the crucible to about 50% capacity.
(e) Place the crucible on a hot plate at 135 + 14°C and evaporate slowly under a fume hood until fuming ceases. Do not allow splattering.
(f) Repeat steps (d) and (e).
(g) Ignite the crucible for 10 minutes under full flame of the burner after the second evaporation.
(h) Cool in a desiccator and weigh the crucible with cover.
(i) Calculate the silica content as follows:

$$\text{Percent Silica Content} = \frac{b - c}{b - a} \times 100$$

Where:
a = tare weight of crucible with cover, gram.
b = initial weight of sample and crucible with cover, gram.
c = final weight of sample and crucible with cover, gram.

4.4.5 Visual examination. The cloth shall be examined for all defects defined in ASTM D3990 and Section 4 which are clearly visible at normal inspection distance (3 feet) shall be scored and assigned demerit. No linear yard (increment of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceed 30 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceed 45 points. If one roll exceeds the point level per 100 square yards, a second sample of 20 rolls shall be examined only for the individual roll quality. If one or more rolls in the second sample of 20 exceeds the individual point value, the lot shall be rejected. Point computation for the lot quality and individual roll quality shall be as follows:

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4.4.5.1 Demerit point computation.

$$\text{Points per 100 square yards} = \frac{3600\ T}{(W \times Y)}$$

Legend:
T = total number of points scored in the sample W = width (in inches) of the cloth purchased
Y = total number of yards inspected

4.4.5.2 Demerit points. Demerit points shall be assigned as follows.
a. For defects up to and including 3 inches in any dimension – one point.
b. For defects exceeding 3 inches but not exceeding 6 inches in any dimension – two points
c. For defects exceeding 6 inches but not exceeding 9 inches in any dimension – three points
d. For defects exceeding 9 inches in any dimension – four points.

The following defects, when present, shall be scored four points for each yard in which they occur.
e. Tear
f. Cut
g. Smash
h. Hole.

4.4.6 Yard-by-yard examination. Each roll in the sample shall be examined as specified in 4.4.1. When the total yardage in the roll does not exceed 100 square yards the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 square yards only 100 square yards shall be examined. Only one roll shall be taken from any shipping container, unless the number of shipping containers in the lot are less than the number of rolls required for sampling, in which case, all shipping containers shall be sampled.

4.4.7 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade when a pigmented cloth is required. Any roll in the sample off- shade, shaded side to side, side to center or end to end shall be cause for rejection of the entire lot presented by the sample (see 3.4 and 6.2).

4.4.8 Width examination. During the yard-by-yard examination, each roll in the sample shall be examined for minimum acceptable width (see 3.7) and selvage width when applicable (see 6.2).

4.4.9 Length examination. During the yard-by-yard examination, each roll in the sample shall be examined for gross length. When total yardage in the roll exceeds 100 yards, only 100 yards shall be examined (see 3.8 and 6.2.).

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4.4.10 Put-up examination. During the yard-by-yard examination, each roll in the sample shall be examined (see 3.8.1).

4.4.11 Fiber identification label examination. During the yard-by-yard examination, each roll in the sample shall be examined for nonconformance. The lot shall be unacceptable if two or more rolls in the sample when the fiber identification label is missing from either or both ends, two or more rolls in the sample are not labeled or ticketed. (see 3.9 and 6.2).

4.4.12 Total yardage in sample examination. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the ticket. The rolls examined shall be those selected for the examination of individual rolls (see 3.8 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The glass cloth is intended for use as protective material during welding and cutting operations.

6.2 Acquisition requirements. Acquisition documents should specify the following:
a. Title, number, and date of this specification.
b. Type, class, National Stock Number.
c. The specific issue of individual documents reference (see Section 2).
d. When first article is required (see Section 3.1, Section 4.2).
e. When required, yard-by-yard examination (see Section 4).
f. Name and address of the first article inspection facility; and the name and the address of the Government activity responsible for conducting the first article inspection program.
g. Inspection conditions.
h. Conformance inspection and acceptable quality levels (see Section 4.3).
i. Packaging required (see Section 5.1).
j. When Certificate of Compliance is applicable.

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6.3 Information requests. For access to information such as patterns, drawings, standard shade samples of cloth etc. go to Defense Supply Center Philadelphia's website for their Specification/Pattern/Drawing Request form, https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx. Complete the request form and then submit. Requests to use equivalent materials and/or components or to make changes to the pattern should be sent to the contracting officer for approval by the military.

6.4 Supersession data. This document supersedes MIL-C-24576A dated 27 July 1987.

6.5 Subject term (key word) listing.

Fire resistant materials
Glass yarn
Noncombustible Oxide,
iron
Oxide, zinc
Silica yarn

6.6 NSN information. The National Stock Numbers (NSNs) associated with the glass cloth are listed below. NSN 8305-01-467-2936 is black on both sides.

NSN	TYPE	CLASS	COLOR	PGC	NSN STATUS
8305-01-035-1056	N/A	N/A	N/A	N/A	Downgraded
8305-01-035-1058	I	1	Tan	11311	Active
8305-01-035-1059	I	2	Tan	11310	Active
8305-01-467-2936	III	1	Black	11308	Active

6.7 Definitions. Sources for definitions cloudy, cut yarn, smash: Fairchild's Dictionary of Textiles, 6th edition, copyright 1979, pages 135, 174, 568.

Sources for definitions cut, hole, tear, The Random House College Dictionary copyright 1984, pages 329, 631, 1348.

Tan.	Defined as a neutral color.
Cloudy.	Defined as thick and thin places in yarn.
Tear.	Defined as to become shredded or separated into
parts Cut.	Defined as to penetrate or divide something.
Cut yarn.	Defined as a defective yarn which is cut or broken in places. The defect is not always apparent, and the yarn may appear sound.
Hole.	Defined as an opening through something, an aperture.
Smash.	Defined as a place in the fabric which a few warp and filling yarns have been broken.

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6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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